On the general twist of type I supergravity

Fridrich Valach

Charles University, Prague

w/ C. Jonker, J. Kupka, I. Saberi, and C. Strickland-Constable

2602.04658, 2604.25803

supported by Charles University grant PRIMUS/25/SCI/018

Supergravity primer

Supergravity: $S(g, \psi, \dots) = \int_M R_g \mathrm{vol}_g + \bar{\psi}_\mu \gamma^{\mu\nu\rho} \nabla_\nu \psi_\rho + \dots$

Supergravity primer

Supergravity: $S(g, \psi, \dots) = \int_M R_g \mathrm{vol}_g + \bar{\psi}_\mu \gamma^{\mu\nu\rho} \nabla_\nu \psi_\rho + \dots$

Symmetries:

- diffeomorphisms $\delta g = \mathcal{L}_x g$, $\delta\psi = \mathcal{L}_x \psi$, $\dots$ (for x a vector field)
- local supersymmetry $\delta g_{\mu\nu} \sim \bar{\epsilon}\gamma_{(\mu}\psi_{\nu)}$, $\delta\psi_\mu = \nabla_\mu \epsilon + \dots$ (for ϵ a spinor)

Supergravity primer

Supergravity: $S(g, \psi, \dots) = \int_M R_g \mathrm{vol}_g + \bar{\psi}_\mu \gamma^{\mu\nu\rho} \nabla_\nu \psi_\rho + \dots$

Symmetries:

- diffeomorphisms $\delta g = \mathcal{L}_x g$, $\delta\psi = \mathcal{L}_x \psi$, $\dots$ (for x a vector field)
- local supersymmetry $\delta g_{\mu\nu} \sim \bar{\epsilon}\gamma_{(\mu}\psi_{\nu)}$, $\delta\psi_\mu = \nabla_\mu \epsilon + \dots$ (for ϵ a spinor)

Symmetry algebra: $[\mathrm{susy}, \mathrm{susy}] = \mathrm{diffeo} + \mathrm{susy} + \mathrm{stuff} \cdot \mathrm{eom}$ $\rightsquigarrow$ for quantisation need BV

Supergravity primer

Supergravity: $S(g,\psi,\dots) = \int_M R_g \mathrm{vol}_g + \bar{\psi}_\mu \gamma^{\mu\nu\rho} \nabla_\nu \psi_\rho + \dots$

Symmetries:

- diffeomorphisms $\delta g = \mathcal{L}_x g$, $\delta\psi = \mathcal{L}_x \psi$, $\dots$ (for x a vector field)
- local supersymmetry $\delta g_{\mu\nu} \sim \bar{\epsilon}\gamma_{(\mu}\psi_{\nu)}$, $\delta\psi_\mu = \nabla_\mu \epsilon + \dots$ (for ϵ a spinor)

Symmetry algebra: $[\text{susy}, \text{susy}] = \text{diffeo} + \text{susy} + \text{stuff} \cdot \text{eom} \quad \rightsquigarrow$ for quantisation need BV

BV formalism:

$$S_{BV}(g,\psi,\dots,\overbrace{\xi, e,\dots}^{\text{ghosts}},\overbrace{g^*,\psi^*,\dots,\xi^*,e^*,\dots}^{\text{antifields}}) = S(g,\psi,\dots) + \int_M g^*(\mathcal{L}_\xi g + \bar{e}\gamma\psi) + \dots + \xi^* \mathcal{L}_\xi \xi + \dots + \psi^*\psi^* \cdot \text{stuff} + \dots$$

such that $\{S_{BV}, S_{BV}\} = 0$

Example: type I/heterotic supergravity [Kupka–Strickland-Constable–FV '25]

Fields: $\underbrace{\text{metric } g\text{, 2-form } B\text{, gauge field } A}_{\mathcal{G}}$, $\underbrace{\text{scalar } \varphi}_{\sigma}$, $\underbrace{\text{gravitino } \Psi_\mu\text{, gaugino } \chi}_{\psi}$, $\underbrace{\text{dilatino } \lambda}_{\rho}$

Example: type I/heterotic supergravity [Kupka–Strickland-Constable–FV '25]

Fields: $\underbrace{\text{metric } g, \text{ 2-form } B, \text{ gauge field } A}_{\mathcal{G}}$, $\underbrace{\text{scalar } \varphi}_{\sigma}$, $\underbrace{\text{gravitino } \Psi_\mu, \text{ gaugino } \chi}_{\psi}$, $\underbrace{\text{dilatino } \lambda}_{\rho}$

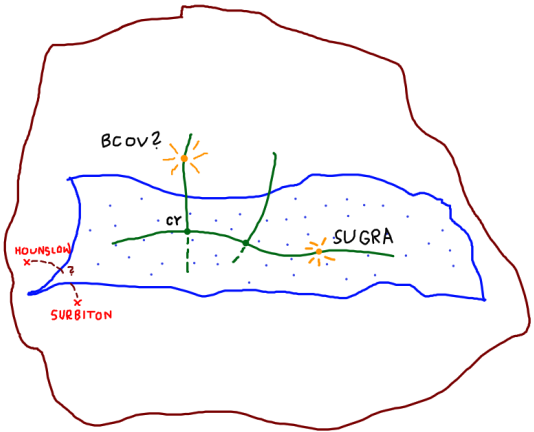
$$
\begin{aligned}
S_{BV} = \int_M \; & \mathcal{R}\sigma^2 + \bar\psi_\alpha \not{D}\psi^\alpha + \bar\rho \not{D}\rho + 2\bar\rho D_\alpha \psi^\alpha \\
& - \tfrac{1}{768}\sigma^{-2}(\bar\psi_\alpha \gamma_{abc}\psi^\alpha)(\bar\rho\gamma^{abc}\rho) - \tfrac{1}{384}\sigma^{-2}(\bar\psi_\alpha\gamma_{abc}\psi^\alpha)(\bar\psi_\beta\gamma^{abc}\psi^\beta) \\
& + \sigma^*[\mathcal{L}_\xi\sigma - \tfrac{1}{8}\sigma^{-1}(\bar\rho e)] + \mathcal{G}^*_{a\beta}[(\mathcal{L}_\xi \mathcal{G})^{a\beta} + \tfrac{1}{2}\sigma^{-2}(\bar e\gamma^a\psi^\beta)] \\
& + \bar\rho^*[\mathcal{L}_\xi\rho + \not{D} e + \tfrac{1}{192}\sigma^{-2}(\bar\psi_\beta\gamma_{abc}\psi^\beta)\gamma^{abc}e] \\
& + \bar\psi^*_\beta[(\mathcal{L}_\xi\psi)^\beta + D^\beta e + \tfrac{1}{8}\sigma^{-2}(\bar\psi^\beta\rho)e - \tfrac{1}{8}\sigma^{-2}(\bar\psi^\beta\gamma_a e)\gamma^a\rho] \\
& + \bar e^*[\mathcal{L}_\xi e + \tfrac{1}{16}\sigma^{-2}(\bar e\gamma_a e)\gamma^a\rho] + \langle \xi^*, Df + \tfrac{1}{2}\mathcal{L}_\xi\xi\rangle - \tfrac{1}{8}\xi^*_a\sigma^{-2}(\bar e\gamma^a e) \\
& + \tfrac{1}{2}f^*(\mathcal{L}_\xi f + \tfrac{1}{8}\sigma^{-2}(\bar e\gamma_a e)\xi^a - \tfrac{1}{6}\langle\xi, \mathcal{L}_\xi\xi\rangle) \\
& - \tfrac{1}{64}\sigma^{-2}(\bar e\gamma_a e)(\bar\psi^*_\beta\gamma^a\psi^{*\beta}) - \tfrac{1}{32}\sigma^{-2}(\bar e\psi^*_\beta)(\bar e\psi^{*\beta}) - \tfrac{1}{64}\sigma^{-2}(\bar e\gamma_a e)(\bar\rho^*\gamma^a\rho^*)
\end{aligned}
$$

Twist of supergravity [Costello–Li '16, '19]

Twist of supergravity = perturbative theory (cyclic L_∞ algebra) obtained by expanding the BV action of supergravity around a critical configuration with nonzero susy ghost

Twist of supergravity [Costello–Li '16, '19]

Twist of supergravity = perturbative theory (cyclic L_∞ algebra) obtained by expanding the BV action of supergravity around a critical configuration with nonzero susy ghost



Twist of type I supergravity

Twist conditions:

metric, 3-form H, scalar φ, principal G-bundle with connection F, spinor e

$$\nabla_\mu e = \tfrac{1}{4} \not{H}_\mu e, \qquad \tfrac{1}{2} \not{H} e = (\not{\nabla} \varphi) e, \qquad \not{F} e = 0, \qquad dH = \mathrm{Tr}(F \wedge F), \qquad e \text{ pure spinor}$$

Twist of type I supergravity

Twist conditions:

metric, 3-form H, scalar φ, principal G-bundle with connection F, spinor e

$$\nabla_\mu e = \tfrac{1}{4} \not{H}_\mu e, \qquad \tfrac{1}{2} \not{H} e = (\not{\nabla} \varphi) e, \qquad \not{F} e = 0, \qquad dH = \mathrm{Tr}(F \wedge F), \qquad e \text{ pure spinor}$$

$\implies$ complex structure, holomorphic volume form Ω, holomorphic principal bundle

Twist of type I supergravity

Twist conditions:

metric, 3-form H, scalar φ, principal G-bundle with connection F, spinor e

$$\nabla_\mu e = \tfrac{1}{4}\not{H}_\mu e, \qquad \tfrac{1}{2}\not{H} e = (\not{\nabla}\varphi)e, \qquad \not{F} e = 0, \qquad dH = \mathrm{Tr}(F \wedge F), \qquad e \text{ pure spinor}$$

$\implies$ complex structure, holomorphic volume form Ω, holomorphic principal bundle

Costello–Li conjecture: the twist on a Kähler Calabi–Yau ($B = \varphi = A = 0$) can be identified with the (minimal) type I BCOV (aka Kodaira–Spencer) theory, i.e. the theory (cyclic L_∞ algebra) describing the deformation of pairs

(complex manifolds, holomorphic volume forms)

Twist of type I supergravity

Twist conditions:

metric, 3-form H, scalar φ, principal G-bundle with connection F, spinor e

$$\nabla_\mu e = \tfrac{1}{4}\not{H}_\mu e, \qquad \tfrac{1}{2}\not{H} e = (\not{\nabla}\varphi)e, \qquad \not{F} e = 0, \qquad dH = \mathrm{Tr}(F \wedge F), \qquad e \text{ pure spinor}$$

$\implies$ complex structure, holomorphic volume form Ω, holomorphic principal bundle

Costello–Li conjecture: the twist on a Kähler Calabi–Yau ($B = \varphi = A = 0$) can be identified with the (minimal) type I BCOV (aka Kodaira–Spencer) theory, i.e. the theory (cyclic L_∞ algebra) describing the deformation of pairs

(complex manifolds, holomorphic volume forms)

Conjecture: the twist in the general case is given by the Courant contact model

Courant contact model [KSSCV '26, JKSSCV '26]

$$\text{Field space} = \Omega^{0,\bullet}[2] \times \Omega^{0,\bullet}(T^{1,0}M \oplus \mathrm{ad}(P_{\mathbb{C}}) \oplus T^{*1,0}M)[1] \times (\Omega^{5,\bullet})^{\times} \ni (f, (x, s, \alpha), \lambda)$$

$$\omega = \delta \int_M \lambda(\delta f + \tfrac{1}{2} i_x \delta\alpha + \tfrac{1}{2} i_{\delta x}\alpha + \tfrac{1}{2}\,\mathrm{Tr}(s\delta s))$$

$$S = \int_M \lambda(L_x^{\partial} f + \tfrac{1}{2} i_x L_x^{\partial}\alpha + \tfrac{1}{2}\,\mathrm{Tr}(s\nabla_x^{\mathbb{C}} s) + \tfrac{1}{6}\,\mathrm{Tr}(s[s,s]) + \bar{\partial} f + \tfrac{1}{2} i_x \bar{\partial}\alpha + \tfrac{1}{2} i_{\bar{\partial} x}\alpha$$
$$+ \tfrac{1}{2}\,\mathrm{Tr}(s\nabla_{0,1}^{\mathbb{C}} s) - \mathrm{Tr}(s i_x F_{\mathbb{C}}^{1,1}) - i_x i_x (i\partial\omega))$$

Courant contact model [KSSCV '26, JKSSCV '26]

$$\text{Field space} = \Omega^{0,\bullet}[2] \times \Omega^{0,\bullet}(T^{1,0}M \oplus \mathrm{ad}(P_{\mathbb{C}}) \oplus T^{*1,0}M)[1] \times (\Omega^{5,\bullet})^{\times} \ni (f, (x, s, \alpha), \lambda)$$

$$\omega = \delta \int_M \lambda(\delta f + \tfrac{1}{2} i_x \delta\alpha + \tfrac{1}{2} i_{\delta x}\alpha + \tfrac{1}{2}\,\mathrm{Tr}(s\delta s))$$

$$S = \int_M \lambda(L_x^{\partial} f + \tfrac{1}{2} i_x L_x^{\partial}\alpha + \tfrac{1}{2}\,\mathrm{Tr}(s\nabla_x^{\mathbb{C}} s) + \tfrac{1}{6}\,\mathrm{Tr}(s[s,s]) + \bar{\partial} f + \tfrac{1}{2} i_x \bar{\partial}\alpha + \tfrac{1}{2} i_{\bar{\partial} x}\alpha$$
$$+ \tfrac{1}{2}\,\mathrm{Tr}(s\nabla_{0,1}^{\mathbb{C}} s) - \mathrm{Tr}(s i_x F_{\mathbb{C}}^{1,1}) - i_x i_x (i\partial\omega))$$

Can be regarded as "resolved" version of [Ashmore–Ibarra–Strickland-Constable–Svanes '25].

Courant contact model [KSSCV '26, JKSSCV '26]

$$\text{Field space} = \Omega^{0,\bullet}[2] \times \Omega^{0,\bullet}(T^{1,0}M \oplus \mathrm{ad}(P_{\mathbb{C}}) \oplus T^{*1,0}M)[1] \times (\Omega^{5,\bullet})^{\times} \ni (f, (x, s, \alpha), \lambda)$$

$$\omega = \delta \int_M \lambda(\delta f + \tfrac{1}{2} i_x \delta\alpha + \tfrac{1}{2} i_{\delta x}\alpha + \tfrac{1}{2}\,\mathrm{Tr}(s\delta s))$$

$$S = \int_M \lambda(L_x^{\partial} f + \tfrac{1}{2} i_x L_x^{\partial}\alpha + \tfrac{1}{2}\,\mathrm{Tr}(s\nabla_x^{\mathbb{C}} s) + \tfrac{1}{6}\,\mathrm{Tr}(s[s,s]) + \bar{\partial} f + \tfrac{1}{2} i_x \bar{\partial}\alpha + \tfrac{1}{2} i_{\bar{\partial} x}\alpha$$
$$+ \tfrac{1}{2}\,\mathrm{Tr}(s\nabla_{0,1}^{\mathbb{C}} s) - \mathrm{Tr}(s i_x F_{\mathbb{C}}^{1,1}) - i_x i_x (i\partial\omega))$$

Can be regarded as "resolved" version of [Ashmore–Ibarra–Strickland-Constable–Svanes '25].

Claim: In the Kähler Calabi–Yau case this recovers the (minimal) type I BCOV theory.

Generalised geometry

Courant algebroid [Liu–Weinstein–Xu '95, Ševera '98]: vector bundle $E \to M$ with fibrewise inner product $\langle \cdot, \cdot \rangle$, bracket $[\cdot, \cdot]$ on sections, and map $\rho \colon E \to TM$, such that Jacobi, ...

Generalised geometry

Courant algebroid [Liu–Weinstein–Xu '95, Ševera '98]: vector bundle $E \to M$ with fibrewise inner product $\langle \cdot, \cdot \rangle$, bracket $[\cdot, \cdot]$ on sections, and map $\rho\colon E \to TM$, such that Jacobi, ...

Courant contact model [KSSCV '26]:

$$\text{Field space} = C^\infty(M)[2] \times \Gamma(E)[1] \times \mathrm{Dens}(M)^\times \ni (f, \xi, \lambda)$$

$$\omega = \delta \int_M \lambda \left(\delta f + \frac{1}{2} \langle \xi, \delta \xi \rangle \right), \qquad S = \int_M \lambda \left(\rho(\xi) f + \frac{1}{6} \langle \xi, [\xi, \xi] \rangle \right)$$

Generalised geometry

Courant algebroid [Liu–Weinstein–Xu '95, Ševera '98]: vector bundle $E \to M$ with fibrewise inner product $\langle \cdot, \cdot \rangle$, bracket $[\cdot, \cdot]$ on sections, and map $\rho\colon E \to TM$, such that Jacobi, ...

Courant contact model [KSSCV '26]:

$$\text{Field space} = C^\infty(M)[2] \times \Gamma(E)[1] \times \mathrm{Dens}(M)^\times \ni (f, \xi, \lambda)$$

$$\omega = \delta \int_M \lambda \left(\delta f + \frac{1}{2} \langle \xi, \delta \xi \rangle \right), \qquad S = \int_M \lambda \left(\rho(\xi) f + \frac{1}{6} \langle \xi, [\xi, \xi] \rangle \right)$$

Generalisation: replace $(C^\infty(M), \Gamma(E))$ by a sheaf of dg rings $\mathcal{O}$ and module $\mathcal{E}$

- $\mathcal{O}$ = holomorphic functions $\rightsquigarrow$ holomorphic Courant algebroids
- $\mathcal{O} = \Omega^{0,\bullet} \rightsquigarrow$ "Dolbeault" Courant algebroids

Geometry of $\mathcal{N} = 1$ supergravity

Setup: transitive Courant algebroid $E = TM \oplus \mathrm{ad}(P) \oplus T^*M$ associated to a principal bundle

Geometry of $\mathcal{N} = 1$ supergravity

Setup: transitive Courant algebroid $E = TM \oplus \mathrm{ad}(P) \oplus T^*M$ associated to a principal bundle

Twist conditions:

$$\bar{e}\gamma_a e = \not{D} e = D_\alpha e = 0$$

Geometry of $\mathcal{N} = 1$ supergravity

Setup: transitive Courant algebroid $E = TM \oplus \mathrm{ad}(P) \oplus T^*M$ associated to a principal bundle

Twist conditions:

$$\bar{e}\gamma_a e = \not{D} e = D_\alpha e = 0$$

$\implies$ $L := \mathrm{annih}(e) \subset E^{\mathbb{C}}$ isotropic involutive

Geometry of $\mathcal{N} = 1$ supergravity

Setup: transitive Courant algebroid $E = TM \oplus \mathrm{ad}(P) \oplus T^*M$ associated to a principal bundle

Twist conditions:

$$\bar{e}\gamma_a e = \not{D} e = D_\alpha e = 0$$

$\implies$ $L := \mathrm{annih}(e) \subset E^{\mathbb{C}}$ isotropic involutive

$\implies$ $L^\perp / L$ is a holomorphic Courant algebroid (reduction)

Geometry of $\mathcal{N} = 1$ supergravity

Setup: transitive Courant algebroid $E = TM \oplus \mathrm{ad}(P) \oplus T^*M$ associated to a principal bundle

Twist conditions:

$$\bar{e}\gamma_a e = \not{D} e = D_\alpha e = 0$$

$\implies$ $L := \mathrm{annih}(e) \subset E^{\mathbb{C}}$ isotropic involutive

$\implies$ $L^\perp / L$ is a holomorphic Courant algebroid (reduction)

$\implies$ $\mathcal{E} := \Omega^{0,\bullet}(L^\perp / L)$ is a Dolbeault Courant algebroid (resolution)

Geometry of $\mathcal{N} = 1$ supergravity

Setup: transitive Courant algebroid $E = TM \oplus \mathrm{ad}(P) \oplus T^*M$ associated to a principal bundle

Twist conditions:

$$\bar{e}\gamma_a e = \not{D} e = D_\alpha e = 0$$

$\implies$ $L := \mathrm{annih}(e) \subset E^{\mathbb{C}}$ isotropic involutive

$\implies$ $L^\perp/L$ is a holomorphic Courant algebroid (reduction)

$\implies$ $\mathcal{E} := \Omega^{0,\bullet}(L^\perp/L)$ is a Dolbeault Courant algebroid (resolution)

Conjecture

The Courant contact model for $\mathcal{E}$ is (a holomorphic realisation of) the supergravity twist.

Outlook

Outlook:

- Understand the deformation problem described by the Courant contact model.
- Prove the Costello–Li conjecture.
- Understand the relation between CCM and the Courant sigma model.
- Build the type II version of the Courant contact model.
- Relation to the deformation theory of generalised torsion-free $U(5) \times O(\dots)$-structures.

Outlook

Outlook:

- Understand the deformation problem described by the Courant contact model.
- Prove the Costello–Li conjecture.
- Understand the relation between CCM and the Courant sigma model.
- Build the type II version of the Courant contact model.
- Relation to the deformation theory of generalised torsion-free $U(5) \times O(\dots)$-structures.

Thank you for your attention!